LLARS: Enabling Domain Experts & Developers Collaboration for LLM Prompting, Generation, and Evaluation

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Abstract

We demonstrate LLARS, an open-source platform that bridges the gap between domain experts and developers for building LLM-based assistants. LLARS integrates three modules into an end-to-end pipeline: **Collaborative Prompt Engineering** for real-time co-authoring with version control and instant LLM testing, **Batch Generation** for systematic output production across prompts × models × data with cost control, and **Evaluation** for human and LLM assessors to rate, rank, label, or compare outputs with live inter-rater agreement metrics. Interviews with six domain experts and three developers in psychosocial counselling confirmed that LLARS eliminates the "translation work" between disciplines and enables domain experts to directly co-author prompts and conduct evaluations independently.

Demo: https://llars-project.github.io/demo

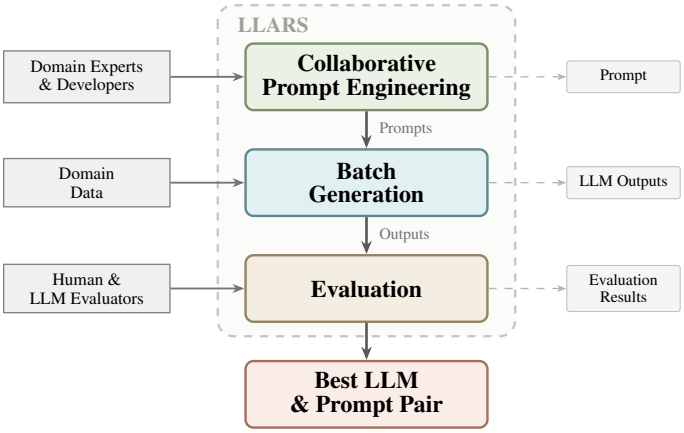

Figure 1: LLARS pipeline: domain experts and developers collaboratively develop prompts, generate outputs across LLMs, and evaluate results with human and LLM evaluators. Each stage supports export; the pipeline yields a validated model–prompt combination.

1 Introduction

Organizations building LLM-based assistants for sensitive domains [Chen *et al.*, 2024; Ling *et al.*, 2025]—psychosocial counseling [Stade *et al.*, 2024], legal analysis [Katz *et al.*, 2024], medical documentation [Van Veen *et al.*, 2024]—must write effective prompts, select suitable models, and rigorously evaluate outputs, requiring both domain expertise (understanding *what* constitutes quality) and technical skill (structuring prompts and configuring LLM pipelines) [Zamfirescu-Pereira *et al.*, 2023; Schulhoff *et al.*, 2024]. Yet these two groups typically work in isolation: prompts are iterated in shared documents without version control, and no single tool supports the full cycle from collaborative prompt development through systematic evaluation.

We present LLARS (LLM Assisted Research System), an open-source platform co-designed with domain experts and developers and iteratively refined through multiple feedback cycles to close this gap with three integrated modules (Figure 1). *Collaborative Prompt Engineering* lets domain experts and developers co-author prompts in a shared real-time editor with version control and instant LLM testing. *Batch Generation* systematically produces the Cartesian product of prompts × models × data items with cost estimation and budget control. *Evaluation* lets human and LLM evaluators assess outputs side by side—via rating, ranking, labeling, or pairwise comparison—with live agreement metrics (Krippendorff's α, Cohen's κ, and others). Each module can be used independently with JSON/CSV export, and the platform connects to any OpenAI-compatible provider, whether commercial or self-hosted.

Table 1 surveys twelve existing platforms along these dimensions. Tracing-oriented tools (Phoenix [Arize AI, 2024], Langfuse [Langfuse GmbH, 2024], W&B Weave [Weights & Biases, 2024]) and prompt managers (Agenta [Agenta AI, 2024], ChainForge [Arawjo *et al.*, 2024]) provide per-output feedback but lack structured human evaluation with agreement metrics. Annotation platforms (Label Studio [HumanSignal, Inc., 2024], Argilla [Argilla, Inc., 2024]) offer structured assessment—Label Studio also computes basic agreement—but require externally generated outputs, disconnecting evaluation from prompt development. Closed-source tools (LangSmith [LangChain, Inc., 2024], Braintrust [Braintrust Data, Inc., 2024], Maxim [Maxim AI, 2024], Vellum [Vellum AI, 2024]) link prompting to evaluation but omit collaborative editing and batch generation as a pipeline

Table 1: Comparison of GUI-first tools for prompt engineering, batch generation, and hybrid LLM output evaluation. ● = native support; ◐ = partial; ○ = not supported. **Collab.** = real-time collaborative prompt co-editing; **Batch** = systematic generation over prompts × models × data; **H-Eval** = structured human evaluation campaigns; **LLM-J** = LLM-as-judge; **IRR** = inter-rater reliability metrics; **E2E** = integrated end-to-end pipeline.

Tool	OSS	Collab.	Batch	H-Eval	LLM-J	IRR	E2E
LLARS	●	●	●	●	●	●	●
Agenta [Agenta AI, 2024]	●	◐	◐	◐	●	○	◐
ChainForge [Arawjo *et al.*, 2024]	●	○	●	○	◐	○	◐
Phoenix [Arize AI, 2024]	●	○	◐	◐	●	○	○
Langfuse [Langfuse GmbH, 2024]	●	○	◐	◐	●	○	○
W&B Weave [Weights & Biases, 2024]	●	○	◐	◐	●	○	○
Label Studio [HumanSignal, Inc., 2024]	●	○	○	●	◐	◐	○
Argilla [Argilla, Inc., 2024]	●	○	○	●	◐	○	○
LangSmith [LangChain, Inc., 2024]	○	○	◐	◐	●	○	◐
Braintrust [Braintrust Data, Inc., 2024]	○	○	◐	◐	●	○	◐
Maxim [Maxim AI, 2024]	○	○	◐	◐	●	○	◐
Vellum [Vellum AI, 2024]	○	○	◐	◐	●	○	◐

stage. LLARS closes this gap as an open-source platform that unifies collaborative prompt engineering, Cartesian-product batch generation, and hybrid human–LLM evaluation with agreement analytics—bringing domain experts and developers together in one end-to-end workflow.

2 Platform Overview

LLARS runs as a containerized web application connecting to any OpenAI-compatible LLM provider, whether commercial or self-hosted. The platform emerged from close collaboration with counselling researchers and practitioners: each module was shaped by recurring usability sessions and revised across multiple development iterations to match real workflow needs. Teams register providers with their own API keys; models then appear throughout the platform.

2.1 Collaborative Prompt Engineering

The prompt engineering module (Figure 2) lets multiple users edit the same prompt simultaneously in a real-time collaborative editor built on Yjs [Jahns, 2023]. Prompts are structured as reorderable message blocks—system instructions, few-shot examples, user templates—with `{{variable}}` placeholders that the platform extracts into a shared palette. For example, a counselling summarisation prompt might contain an instruction block ("Write a situation overview in 2–3 sentences"), a format specification, and a data block with `{{subject}}` and `{{content}}` placeholders marking exactly where client data will be inserted. This separation keeps the prompt logic readable—domain experts see the instruction structure at a glance rather than buried in pasted text—and allows team members to assign different sample values to each variable (e.g. swapping a custody dispute for a school-refusal case) to verify that the prompt generalises rather than overfitting to a single example. The same variable names carry over to batch generation, where `{{subject}}` and `{{content}}` map to identically named columns in the uploaded dataset; the prompt developed during engineering thus processes an entire corpus without modification, keeping

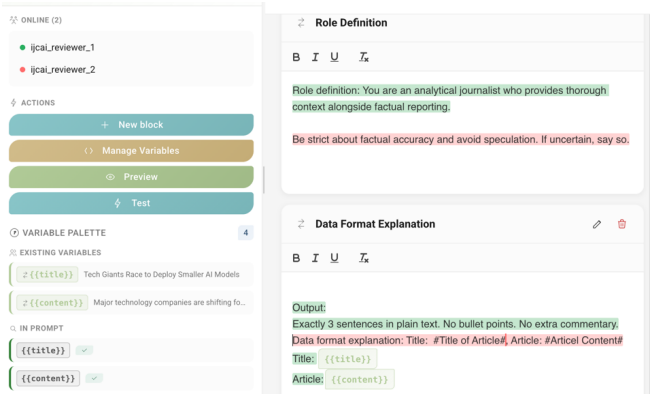

Figure 2: Collaborative prompt editor showing a news summarization prompt with two message blocks, template variables in the shared palette (left), and online presence. The sidebar provides testing, variable management, and JSON import/export.

the development workflow close to production use. Any collaborator can test the current prompt against a selected LLM while editing continues, and multiple team members can test different models or sample data in parallel. A version control system tracks every revision with multi-author attribution, supporting comparison and rollback. Finalized prompts are exported as JSON or passed directly to batch generation.

2.2 Batch Generation

Once prompts are refined, the batch generation module scales them to full datasets. Consider a research team that has collaboratively developed two candidate prompts—one producing structured bullet-point analyses, the other generating narrative summaries—and wants to compare them across a self-hosted and a commercial model. A wizard guides users through three data source modes, each suited to a different stage of development.

In *new-data mode*, users upload a CSV or JSON file whose columns map to the prompt's template variables. The team

uploads 50 counselling email threads as JSON, each entry containing `subject` and `content` fields matching the variables defined during prompt engineering. LLARS computes the full Cartesian product: 50 threads $\times$ 2 prompts $\times$ 2 models = 200 outputs. Before execution, a generation matrix previews this combination and estimates cost based on model pricing and projected token counts; optional budget caps pause jobs that exceed a threshold.

If the team later adds a third model, *previous-generation mode* lets them re-run the same 50 threads with the new model—producing 100 additional outputs that join the existing provenance chain without re-uploading data. This is also useful when the team refines a prompt after reviewing initial outputs: the revised prompt can be applied to the original dataset in a single step.

Prompt-only mode serves a different purpose. Before investing in data preparation, teams send multiple prompts to multiple models without any external data—for instance, testing a structured versus a narrative system prompt across three candidate models to compare output style and quality, yielding six responses to review rather than hundreds. This lightweight comparison helps narrow the search space before committing to a full data-intensive run.

Results stream in real time, and every output records its full provenance—source item, prompt version, model, parameters, token count, and cost—exportable as CSV or JSON. This provenance is central to the pipeline: when outputs reach evaluation, assessors see only blinded labels, and the platform maps their judgments back to specific model–prompt combinations, directly informing selection. Completed batches transfer into evaluation with a single click, preserving all attribution.

2.3 Evaluation

The evaluation module brings human and LLM evaluators together in a single framework. Four evaluation types are available (Figure 3): *rating* with configurable multi-dimensional Likert scales, *ranking* into ordered buckets, *labeling* for binary or multi-class classification, and pairwise *comparison*; the type system is extensible (Section 3). An AI-assisted Scenario Wizard analyzes uploaded data to recommend the most suitable type; scenarios can also be created directly from batch generation outputs, preserving model attribution.

Scenario owners control how items are distributed—all items to every evaluator for full inter-rater analysis, or configurable subsets for faster throughput. Where most tools limit LLMs to pairwise judging [Zheng *et al.*, 2024], LLARS treats them as full evaluation participants that rate, rank, label, and compare alongside humans using customizable type-specific prompts. Agreement metrics update live—Krippendorff's α, Cohen's and Fleiss' κ, Kendall's τ, and ICC—filterable by evaluator pairing (human–human, LLM–LLM, human–LLM). All data, metrics, and configurations are exportable as JSON or CSV.

3 Deployment and User Study

LLARS is actively used in several research projects in psychosocial counselling and beyond. The Virtual Client [Rudolph *et al.*, 2024] leverages LLARS to develop and evaluate prompts that power generative-AI-based simulations of client interactions for counsellor training. CAIA [Steigerwald *et al.*, 2025] is a co-designed AI assistant for text-based online counselling that augments professional practice through seven LLM-driven functionalities—extracting, summarising, and interpreting client communications alongside the traditional email exchange. One of CAIA's functionalities, automated subject line generation, served as a full end-to-end case study [Steigerwald and Albrecht, 2025]: prompts were collaboratively refined in LLARS, batch generation produced 253 subject lines across 11 LLMs of varying size, and five professionals together with an LLM evaluator rated them (1,518 assessments)—determining how small a model could be while still producing meaningful results. LLARS also motivated two domain-specific evaluation types (Figure 3): *Mail Rating* and *Authenticity*.

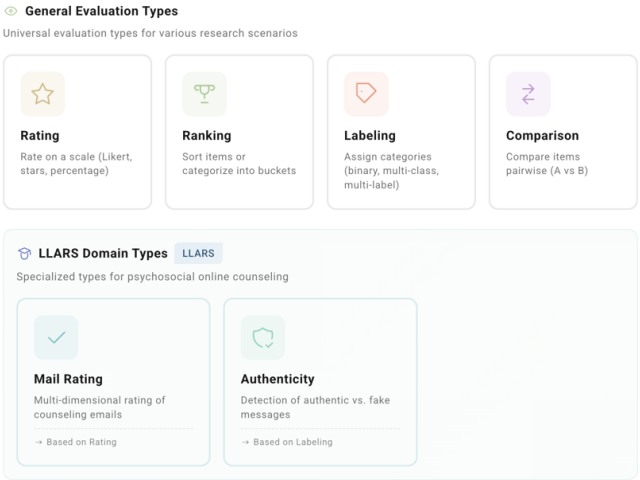

Figure 3: AI-assisted Scenario Wizard showing the task selection step with four general evaluation types and two domain-specific types for psychosocial counseling. The five-step wizard analyzes uploaded data and recommends the most suitable evaluation type.

Semi-structured interviews with six domain experts and three developers confirmed that consolidating the full pipeline in a single platform saves considerable time. Regarding *prompt engineering*, domain experts found that the collaborative editor made prompt authoring straightforward—particularly when working with self-hosted models, where no external tooling or playground exists. For *batch generation*, developers valued that prompts, data, and outputs remain in one place, eliminating the manual export–import cycles between separate tools that previously slowed iteration. The *evaluation* module was rated as immediately understandable by all domain experts: clear progress indicators show which items have been assessed and which remain, letting evaluators resume across sessions without losing context. Across all three modules, both groups emphasised that co-locating the entire workflow removed the "translation work" between disciplines and created the experience of "truly working together on the same task."

4 Limitations, Demo, and Future Work

LLARS has so far been deployed in a single domain with a small user base, so broader studies are needed to assess generalisability. The quality of LLM evaluator judgments depends on careful prompt design, and the current evaluation relies on qualitative interviews rather than a controlled study. While the variable system helps prevent overfitting during prompt development, practitioners noted that prompts optimised on test cases can still underperform on unseen real-world data—a limitation inherent to any iterative prompt engineering process.

Our live demo walks attendees through the full pipeline, from collaborative prompt editing through batch generation to evaluation with live metrics. Future work targets a controlled user study comparing LLARS against baseline workflows (shared documents and spreadsheets) and cross-domain validation in legal and medical settings. LLARS is open-source and under active development; it continues to be deployed in ongoing projects to bring domain experts and developers together so that diverse fields beyond counselling can benefit from the current advances in AI.

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